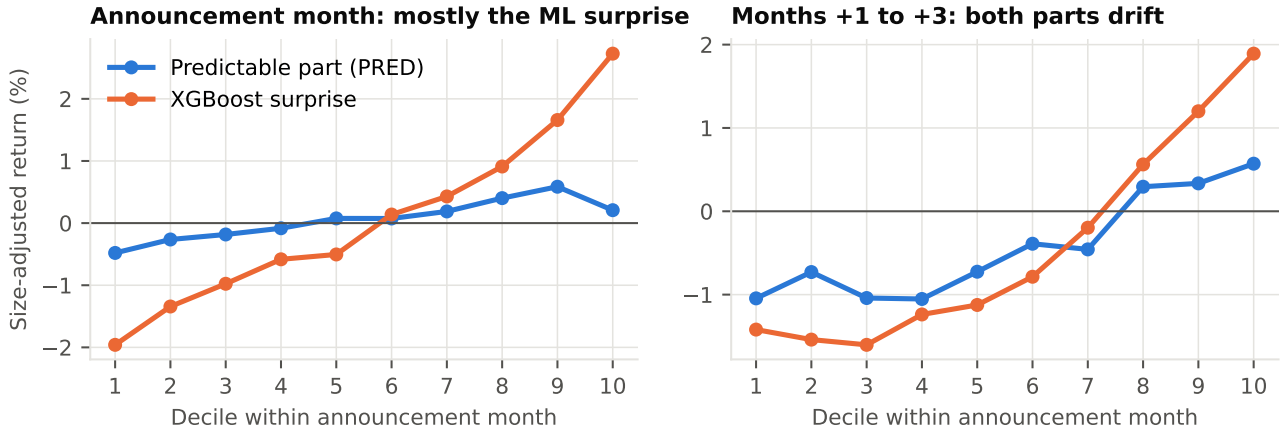
Idea 4 PoC — Whose earnings expectations do Chinese prices reflect?

Proof of concept, October 2026. Code: `idea04_whose_expectations/run.py`.

Question and design

Do A-share prices treat earnings as surprising relative to a naive seasonal random walk (SRW), or relative to a better, look-ahead-free forecast? The SRW surprise splits exactly into two parts. The first is the part that the out-of-sample XGBoost model predicted before the announcement, $PRED = (\hat{E}^{XGB} - \hat{E}^{SRW})/\sigma$. The second is the XGBoost surprise $SUE^{XGB} = (E - \hat{E}^{XGB})/\sigma$. Both share the same scale σ, the SD of the firm's last eight SRW errors. If the market used naive expectations, $PRED$ would move announcement returns as much as SUE^{XGB} does (as in Bernard and Thomas, 1990). If it used ML-quality expectations, $PRED$ would already be priced. Sample: 104,487 quarterly announcements, 2005–2020, 158 announcement months. Returns are adjusted with size-decile benchmarks. Estimation is Fama–MacBeth by announcement month with Newey–West t-statistics.



Window (size-adjusted return)	Months $-2, -1$	Announcement month	Months $+1$ to $+3$
$PRED$ (predicted before announcement)	1.08 (6.6)	0.11 (0.9)	0.60 (3.5)
SUE^{XGB} (XGBoost surprise)	0.97 (8.4)	0.73 (9.7)	0.52 (4.7)
Ratio $PRED/SUE^{XGB}$	1.11	0.15	1.16

Coefficients in % return per unit of SUE (t-statistics). Announcement-month test of equality: $t = -5.8$. With size, B/M, momentum and turnover controls, the announcement-month $PRED$ coefficient is 0.18 ($t = 1.4$).

Findings

- **At the announcement, prices react to the ML surprise, not the predictable part.** The announcement-month response to $PRED$ is only 15% of the response to SUE^{XGB} and is statistically zero. The market does not act as if it used the naive SRW benchmark. The two surprises explain about the same share of returns (R^2 2.0% against 1.9%).
- **The predictable part is priced slowly, not at the announcement.** $PRED$ predicts returns in the two months *before* the announcement, consistent with leakage and China's mandatory preannouncements. It also predicts returns in the three months *after*, by as much as the true surprise does. Information that a statistical model had in advance is still being incorporated one to three months after it is disclosed.
- **Analyst coverage behaves unexpectedly.** The announcement-month reaction to $PRED$ is zero for firms without analyst coverage, but positive for covered firms (0.42, $t = 2.9$). A possible reason is that analysts' forecasts themselves lag the statistical forecast.
- **A tradable version is weak.** Sorting announcers at the start of the month on $PRED$, using only inputs public before the month, earns a Q5−Q1 spread of 0.45%/month ($t = 1.6$; CH-4 $\alpha = 0.45\%$, $t = 1.5$).

Assessment and next steps

The headline is not "naive expectations". It is a timing result: predictable earnings information is priced gradually around the announcement rather than at it. That is a sharper and more novel framing. Next steps: (i) use daily returns to separate the preannouncement, announcement-day and drift windows; (ii) merge preannouncement records to test whether the pre-announcement $PRED$ effect is the preannouncement channel; (iii) add an analyst quarterly surprise, which needs I/B/E/S quarterly detail rather than the 12-month blend in the panel; (iv) run placebo tests with shuffled $PRED$.